STACKS, DERIVED CATEGORIES OF SPACES: EXACT-SEQUENCE-LOWER-SHRIEK

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1822-1842 of sources/stacks-spaces-perfect/pinned-original-source.tex; proof span is lines 1844-1886, the complete author proof (41 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 8 of queue_spaceperfect.json (pb=1661, stmt 1822-1842, proof 1844-1886).

Editorial note. stacks-spaces-perfect wave sp01 (2026-10-14): fresh adjudication accept per lane-s3/sp01_judgments; pool per Chief 12:4x unfreeze (reserve-pool workflow, six-check clean); spaces-perfect.tex 126 with-proof rows, 70 b107-occupied labels avoided, 56 free rows judged 32 accept / 24 reject.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-exact-sequence-lower-shriek' also exists in stacks-cohomology — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0152: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1886–1886. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 014Z: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1886–1886. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08GU: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1810–1813. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08GT: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1799–1801. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08FH: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1886–1886. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04K5: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1886–1886. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $(U \subset X, V \to X)$ be an elementary distinguished square of algebraic spaces over S.*

(1) *For a sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$ we have a short exact sequence*
$$0 \to j_{U \times_X V!}\mathcal{F}|_{U \times_X V} \to j_{U!}\mathcal{F}|_U \oplus j_{V!}\mathcal{F}|_V \to \mathcal{F} \to 0$$
(2) *For an object E of $D(\mathcal{O}_X)$ we have a distinguished triangle*
$$j_{U \times_X V!}E|_{U \times_X V} \to j_{U!}E|_U \oplus j_{V!}E|_V \to E \to j_{U \times_X V!}E|_{U \times_X V}[1]$$
in $D(\mathcal{O}_X)$.

Proof. To show the sequence of (1) is exact we may check on stalks at geometric points by Properties of Spaces, Theorem 04K5. Let $\overline{x}$ be a geometric point of X. By Equations (08GT) and (08GU) taking stalks at $\overline{x}$ we obtain the sequence
$$0 \to \bigoplus\nolimits_{(\overline{u},\overline{v})} \mathcal{F}_{\overline{x}} \to \bigoplus\nolimits_{\overline{u}} \mathcal{F}_{\overline{x}} \oplus \bigoplus\nolimits_{\overline{v}} \mathcal{F}_{\overline{x}} \to \mathcal{F}_{\overline{x}} \to 0$$
This sequence is exact because for every $\overline{x}$ there either is exactly one $\overline{u}$ mapping to $\overline{x}$, or there is no $\overline{u}$ and exactly one $\overline{v}$ mapping to $\overline{x}$.

Proof of (2). We have seen in Cohomology on Sites, Section 08FH that the restriction functors and the extension by zero functors on derived categories are computed by just applying the functor to any complex. Let $\mathcal{E}^\bullet$ be a complex of $\mathcal{O}_X$-modules representing E. The distinguished triangle of the lemma is the distinguished triangle associated (by Derived Categories, Section 014Z and especially Lemma 0152) to the short exact sequence of complexes of $\mathcal{O}_X$-modules
$$0 \to j_{U \times_X V!}\mathcal{E}^\bullet|_{U \times_X V} \to j_{U!}\mathcal{E}^\bullet|_U \oplus j_{V!}\mathcal{E}^\bullet|_V \to \mathcal{E}^\bullet \to 0$$
which is short exact by (1). □